Magnetic Effects of Electric Current

Science • Chapter 12 • *Chapter Summary & Study Guide for Daksh*

An electric current produces a magnetic effect — a discovery (by Hans Christian Oersted in 1820) that linked two distinct phenomena and gave us motors, generators, transformers, MRI scanners and electromagnets. The reverse is also true (a changing magnetic field produces a current — Faraday's discovery). This chapter explores both effects and their everyday applications.

Magnetic field — what is it?

A **magnetic field** is the region around a magnet (or current-carrying conductor) where its magnetic effect can be felt. It is represented by **field lines**, which point from N-pole to S-pole outside a magnet (and S to N inside). Field lines never intersect (a single direction of B at every point); their density indicates field strength. The simple compass needle is a small magnet that aligns with the field.

Magnetic field due to current-carrying conductors

- **Straight wire:** field lines are concentric circles around the wire. *Right-hand thumb rule*: if you grasp the wire with the right hand, thumb pointing along current, fingers curl in the direction of B.
- **Circular loop:** field lines form concentric circles near the wire that combine to give a roughly straight field through the centre, perpendicular to the loop's plane.
- **Solenoid** (long coil): field inside is uniform and strong, like that of a bar magnet — making one end an N-pole and the other an S-pole. The field strength can be greatly increased by inserting a soft iron core — this is an **electromagnet**.

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Right-hand thumb rule:        Fleming's left-hand rule (force on a current
                              -carrying conductor in a magnetic field):
  Thumb : current               Forefinger -> Field (B)
  Fingers : magnetic field      seCond     -> Current  (I)
                                Thumb      -> Thrust / Motion (F)
Mnemonic: 'F-B-I'
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Force on a current-carrying conductor in a magnetic field

When a current-carrying conductor is placed in a magnetic field, it experiences a force perpendicular to both. The direction is given by **Fleming's left-hand rule**. The force is maximum when current is perpendicular to the field, zero when parallel. This is the principle of an **electric motor**:

- A coil ABCD between the poles of a magnet, connected to a battery via brushes B_1, B_2 and a **split-ring commutator**.
- Current produces equal and opposite forces on AB and CD → couple → coil rotates.
- The split-ring reverses current every half-rotation, ensuring continuous rotation in the same direction.
- Motors are everywhere — fans, washing machines, blenders, vehicles, pumps, drills.

Electromagnetic induction

Faraday discovered that whenever the magnetic flux linked with a coil changes, an EMF (and therefore a current in a closed circuit) is induced in the coil. This is **electromagnetic induction**. Direction is given by **Fleming's right-hand rule**. This is the principle of an **electric generator** — a coil rotated in a magnetic field produces alternating current. Power stations use this to convert mechanical energy (steam/water/wind-driven turbines) into electrical energy.

AC vs DC

- **AC (alternating current)** reverses direction periodically (50 Hz in India). Easier to transmit over long distances using transformers.
- **DC (direct current)** flows in one direction (cells, batteries).

Domestic electric circuits & safety

Power reaches our homes via three wires — **live** (red, 220 V), **neutral** (black, 0 V), and **earth** (green, connected to ground). All appliances are in parallel. **Short-circuit** occurs when live and neutral touch directly — current shoots up; **overloading** occurs when too many appliances draw current — wires heat up. **Fuses** and **MCBs** (miniature circuit breakers) protect against both. **Earthing** protects users from electric shock if the appliance body becomes live.

★ MUST-REMEMBER POINTS

- Right-hand thumb rule (current $\leftrightarrow$ field around a wire); Fleming's left-hand rule (force on current-conductor: F-B-I).
- Solenoid acts like a bar magnet; with iron core, becomes an electromagnet.
- Electric motor converts electrical $\rightarrow$ mechanical energy; uses split-ring commutator.
- Faraday's law: changing magnetic flux induces an EMF (basis of electric generator).
- AC frequency in India = 50 Hz; AC easier to transmit (transformers).
- Three wires in domestic supply: live, neutral, earth. Earthing prevents shock.
- Fuse/MCB protects against overload & short circuit (heating effect).

— End of Summary —